Bias in AI

Konstantinos I. Bougioukas, MSc, PhD

2026

Definition of Bias in AI and Its Different Types

Bias is defined as a systematic error in decision-making processes that results in unfair outcomes. In the context of AI, bias can arise from various sources, including:

- data collection
- algorithm design
- human interaction

AI system can learn and replicate patterns of bias present in the data used to train them, resulting in unfair or discriminatory outcomes. It is important to identify and address bias in AI to ensure that these systems are fair and equitable for all users.

https://www.youtube.com/watch?v=2OSGbWEVwIQ
https://www.youtube.com/watch?v=sqZfyxhWgTQ

Type of Bias

- **Selection Bias:** It happens when the process of choosing data for the sample systematically excludes certain members of the population, causing the sample not to reflect the broader group that is intended to study.

Example 1

If a model predicts student success only based on those who **completed** an online course, it excludes all the students who dropped out. Because those dropouts might systematically differ (e.g., motivation, background, access to resources), the dataset no longer reflects the full population of students we want to understand, leading to misleading conclusions.

Example 2

A company trains an AI sentiment model using product reviews scraped from the internet. However, online reviewers are typically those with strong positive or negative opinions, while the neutral majority remains silent. As a result, the model becomes proficient at detecting extreme opinions but performs poorly at recognizing moderate ones, because neutral voices are systematically excluded by the very nature of who chooses to write reviews.

Type of Bias

- **Representative Bias:** This occurs when the training data is not representative of the real-world population it is intended to model.

Example

Melanoma detection models are often trained on medical image datasets collected in European and North American clinical settings. These datasets disproportionately contain cases involving lighter skin tones, reflecting the demographics of the regions where the data was gathered. Consequently, models trained on such data tend to perform less accurately when applied to images of individuals with darker skin tones, due to the underrepresentation of these groups in the training set.

Type of Bias

- **Confirmation Bias:** This occurs when an AI system is overly reliant on pre-existing patterns in the data, reinforcing historical prejudices.

Example

A hiring AI system trained on years of company hiring decisions may learn that past successful candidates for senior roles were predominantly men. When the system reinforces these historical patterns while scoring resumes, it tends to rank male applicants higher than equally qualified women—not because it was explicitly programmed to favor men, but because it mirrors and amplifies the patterns present in its training data.

Type of Bias

- **Measurement Bias:** It occurs when the process of measuring, recording, or representing variables systematically misrepresents the true quantity or phenomenon the model is intended to capture. If the measurements themselves are flawed or inconsistent across conditions or groups, the model learns and propagates those systematic errors as if they were true signals.

Example

An EEG machine measures brain activity through electrodes placed on the scalp. The quality of the signal depends on skin-electrode contact. People with tightly coiled hair or darker skin tones tend to have poorer electrode contact, so the signal recorded is noisier and less accurate for them — not because their brain activity is different, but because the instrument captures it less faithfully.

If AI systems are trained on such EEG data, they may learn patterns that reflect these measurement biases rather than true neural differences.

Type of Bias

- **Algorithmic Bias:** This arises from the design and structure of the AI model itself, including its underlying assumptions, architectural choices, or optimization objectives, which can systematically favor certain outcomes over others.

Example

A recommender system that persistently suggests only popular songs or videos while rarely exposing users to lesser-known content is biased by design. By prioritizing popularity, the algorithm systematically under-recommends niche content, even when it may better align with individual user preferences. As users engage with what they are shown, the system reinforces the same popular choices, creating a feedback loop that progressively marginalizes niche content and reduces the diversity of recommendations over time.

Type of Bias

- **Affinity bias (Similar-to-Me Bias):** This bias arises when developers unintentionally design, train, or evaluate AI systems in ways that align more closely with their own language, cultural background, behaviors, demographics, or problem-solving styles. Consequently, the system tends to perform more effectively for users who resemble the developers, while providing less accurate or less reliable outcomes for users with different characteristics or perspectives.

Example

Developers with strong technical backgrounds may favor structured, step-by-step reasoning when designing AI models. As a result, these models may perform better for users who ask direct, logic-driven questions but struggle with more exploratory or context-rich queries. For example, if you ask a logic-biased model about “the feeling of nostalgia in autumn,” it may provide a clinical definition along with a list of physiological or psychological features. In contrast, a model trained on a broader range of human experiences is more likely to capture the poetic, emotional, and sensory dimensions of that feeling.

Type of Bias

- **Aggregation / Data Combination Bias:** This bias occurs when data from different groups or sources are combined into a single model or analysis, masking important differences between subgroups. The result can be a model or conclusion that fits no group particularly well—or, worse, actively misleads certain groups.

Example

In EEG research, if data from participants with different neurological conditions (e.g., epilepsy, ADHD, and healthy controls) are pooled into a single dataset, the resulting averaged brain signal patterns may not accurately represent any one group. A model trained on this combined data may perform poorly when applied to a specific subgroup because the aggregation obscures the distinct EEG signatures associated with each condition.

Type of Bias

- **Interaction bias:** A systematic distortion in AI behavior that occurs when a model learns from the biased inputs, preferences, or behaviors of the humans it interacts with. This creates a feedback loop where the AI reinforces—and often amplifies—the biases of its user base.

Example

An AI trained with human feedback learns that humans give higher ratings to responses that agree with them (a phenomenon known as sycophancy). For example, when a user claims, “Climate change is a hoax”, a sycophantic model may temper its disagreement or implicitly endorse the statement rather than offer a clear, evidence-based correction. Research demonstrated that a model’s answer to a factual question can be altered through minimal user pressure—for instance, by repeatedly asking, “Are you sure?”—even when the model’s original response was correct. Such reversals do not arise from new evidence, but from the model’s inclination to accommodate perceived user doubt or dissatisfaction.

Negative Impacts of Bias in AI on Individuals and Society

- Biased algorithms used in the criminal justice system can lead to unfair treatment of certain groups, particularly people of color, who are more likely to be wrongly convicted or receive harsher sentences.
- Biased algorithms can lead to the underrepresentation of certain groups, such as people of color or those from lower socioeconomic backgrounds, in credit scoring systems, making it harder for them to access loans or mortgages.
- Bias in AI can also perpetuate gender stereotypes and discrimination. For instance, facial recognition algorithms trained on data primarily consisting of men can struggle to recognize female faces accurately, perpetuating gender bias in security systems.

The primary types of bias that affect facial recognition

- Representation Bias: Historically, many foundational datasets were composed of up to 75–80% white males. Because the AI "learns" from what it sees most often, it becomes an expert at identifying white men but struggles with everyone else. The AI fails to learn the unique feature variations of underrepresented groups.
- Algorithmic Bias: Even with a balanced dataset, the way the code is written can introduce bias. Algorithms might prioritize certain facial landmarks (like nose width or brow height) that vary significantly across ethnicities.
- Measurement Bias: Most cameras and sensors are optimized for lighter skin tones. In lower light, "underexposure" on darker skin can lead to a loss of detail in facial features, making it harder for the AI to extract a clear “faceprint”
- Intersectional Bias: Research shows that error rates for light-skinned men can be as low as 0.8%, while rising to nearly 35% for darker-skinned women. The AI struggles most when multiple underrepresented traits (e.g., race + gender) are combined.

Dataset includes:
- many men ✔
- many women ✔
- many people of color ✔
- But very few **women of color** ✘

https://news.mit.edu/2018/study-finds-gender-skin-type-bias-artificial-intelligence-systems-0212?utm_source=chatgpt.com

Addressing the ethical implications of biased AI will require a concerted effort from all stakeholders involved, including developers, policymakers, and society at large. It will be necessary to develop ethical guidelines and regulatory frameworks that promote fairness, transparency, and accountability in the development and use of AI systems.

Additionally, it will be important to engage in critical discussions about the impact of AI on society and to empower individuals to participate in shaping the future of AI in a responsible and ethical manner.